Project Plan

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idea:

App that helps students and research teams at TU Wien work more effectively with heterogeneous knowledge (notes, PDFs, slides, recordings, screenshots). It ingests multi-format material, enriches it with metadata, enables hybrid search and connection discovery, and generates task-ready outputs (summaries, citations, outlines). The prototype focuses on high-quality retrieval, trustworthy summarisation, and proactive surfacing of relevant context while writing or studying.

1. Users

Our primary users are students and early-career researchers at TU Wien who work with large amounts of study materials, papers, and project documentation across multiple courses and research topics. These users often collect information from many different sources, but lack a unified way to organize or retrieve it efficiently.

2. Data

Our target users, students and research groups at TU Wien, work with a wide range of study and research materials that primarily live on their personal devices. This includes structured content such as spreadsheets or exported tables, but the majority of their information is contained in semi-structured or unstructured formats: PDF research papers, PowerPoint lecture slides, Markdown notes, Word documents, images of handwritten notes, screenshots, and audio recordings from lectures or meetings. Each file is accompanied by basic metadata such as timestamps, file extensions, and folder locations; however, this metadata is often incomplete or inconsistent across sources.

Granularity: The materials vary significantly in granularity. Some files are highly atomic—for example, a short Markdown note summarizing one concept—while others are long, dense documents covering multiple unrelated topics, such as 40-page papers or multi-chapter lecture slide decks. Because the system must operate across this full spectrum, it must extract fine-grained semantic units even from large, multi-topic documents.

Connections: In most cases, relationships between pieces of information are implicit rather than explicit. Students rarely maintain structured linking systems, and different materials

rarely reference each other directly. The system must therefore infer connections through semantic similarity, cross-document topic extraction, and embedded metadata where available.

Completeness: The completeness of the materials varies widely. Some research papers are polished and self-contained, whereas many personal notes are fragmented, incomplete, or written in shorthand. Audio recordings may include noise or partial context. This inconsistency requires robust preprocessing to make the materials usable for downstream search and recommendation.

Context: Context exists but is uneven. Metadata such as creation dates, file names, author names in PDFs, or the folder in which a file is stored can give hints about when or why a piece of content was created. However, users' organizational habits differ widely, and context is often missing or ambiguous. To address this, the system augments metadata with extracted information such as document titles, section headers, timestamps from transcripts, and topic summaries to reconstruct context more reliably.

Heterogenity: The dataset is highly heterogeneous. Materials come in different file formats, follow different conventions, and vary in structure, language, and quality. Some PDFs contain selectable text while others require OCR; handwritten notes may be photographed at low resolution; audio recordings may differ in clarity and accent; and slide decks often include diagrams that require multimodal processing. This heterogeneity is central to the challenge of building a unified search system.

3. The Problem

The core challenge our target users - students and research teams at TU Wien - face is knowledge fragmentation and synthesis friction. Their academic and research workflows require them to constantly acquire, process, and create complex information. However, their digital knowledge assets are scattered across a highly heterogeneous landscape - from PDFs and slides to audio recordings, screenshots, and markdown notes. This fragmentation creates significant, tangible problems.

The first issue is inefficient and failed retrieval. Users struggle with the "I know I saw this somewhere" problem because standard keyword search is ineffective. It cannot find a concept discussed in an audio recording, visually represented on a screenshot, or semantically related to a term in a PDF. A student searching for "recursive neural networks" might find their lecture notes but completely miss the foundational concepts in a PDF paper or the practical details on a slide screenshot. This forces users to rely on their memory of where and in what format information exists, rather than just what the information is.

Beyond simple retrieval, this siloed structure causes missed connections and poor synthesis. The most significant insights often lie at the intersection of different knowledge pieces, but because the data is disconnected, these connections remain implicit and undiscovered. A student may fail to see how a concept from a "Machine Learning" lecture (.pptx) directly applies to a problem in their "Signal Processing" seminar (.md notes). This inability to

"connect the dots" hinders deep understanding, prevents novel idea generation, and makes "seeing the big picture" nearly impossible.

This combination of poor retrieval and missed connections results in stale knowledge and a high cognitive load. Information gets buried; a crucial insight from a first-semester course is never seen again when it becomes relevant during a final-year thesis. When performing a task like writing a literature review, users must expend enormous cognitive effort to manually find, re-read, compare, and synthesize information from dozens of different sources and formats. This "input friction" for creating new, synthesized knowledge is a major bottleneck to productivity and innovation.

4. The Solution

Concept

The system will ingest heterogeneous knowledge materials from students and research teams at TU Wien, including personal notes, PDF papers, lecture slides, audio recordings, and screenshots. Upon ingestion, the system automatically enriches each artifact with metadata such as creation timestamps, file types, and extracted topics to facilitate organization and retrieval. The core functionalities include:

Understand: The system parses diverse file formats and extracts structured information, identifying key topics, important terms, entities, and main concepts to enable effective downstream search and analysis.

Retrieve: The system implements hybrid search, combining keyword-based retrieval for exact term matching with semantic search using dense vector embeddings to capture contextual meaning. This dual approach enables users to find both precise references and conceptually related materials, balancing specificity with semantic relevance.

Connect: The system automatically identifies relationships between materials based on shared entities, overlapping topics, and semantic similarity. By surfacing implicit links—such as connections between lecture slides and related PDF papers or notes—the system helps users discover related materials they may have overlooked.

Generate: The system produces task-ready outputs tailored to academic workflows, including concise summaries of lengthy documents, properly formatted citations for academic writing, and structured study outlines. These outputs reduce cognitive load and accelerate knowledge work.

Surface: The system proactively recommends relevant materials based on the user's current context—their active work, recent searches, or document being

edited—without requiring explicit queries. For instance, while drafting a report on cloud computing, the system automatically surfaces related notes, relevant citations, or lecture recordings on the topic.

User Interface

The system will be delivered as a cross-platform desktop application compatible with Windows, macOS, and Linux. This interface choice supports offline access to personal knowledge bases, which is critical for students and researchers working in environments with limited connectivity or handling sensitive data. A desktop application also enables deeper integration with local file systems, facilitating seamless ingestion of materials stored across diverse folders and drives. Users will interact through a streamlined search interface, a document viewer with inline connection visualization, and a sidebar for context-aware recommendations.

Technical Approach

Three core technical challenges will guide system development:

Hybrid retrieval optimization: Balancing semantic and keyword search requires tuning the relative weights of sparse vectors (exact matching) and dense vectors (contextual understanding) using techniques such as reciprocal rank fusion. Testing different embedding models and chunking strategies will be essential to achieve high precision and recall across diverse material types.

Multi-format ingestion and enrichment: Extracting meaningful text and metadata from PDFs, slides, handwritten notes, and audio recordings involves optical character recognition (OCR), speech-to-text transcription, and layout parsing. Ensuring consistent quality and completeness across formats represents a key technical challenge, particularly for noisy or incomplete inputs.

Context-aware proactive surfacing: Identifying materials relevant to a user's current activity without explicit queries requires real-time analysis of document context, topic detection, and relevance ranking based on both content similarity and recency. Building a lightweight, responsive recommendation engine that enhances rather than disrupts user workflows is critical for system usability.

5. Evaluation

The proposed system offers a compelling solution to a real and well-documented problem among students and research teams at TU Wien: fragmented, heterogeneous knowledge

that is difficult to retrieve, connect, and synthesize. Overall, the concept is feasible, highly relevant to the workflows of the target users, and technically ambitious in meaningful ways. This evaluation considers the strengths of the approach, its technical and practical challenges, and the criteria by which its success can be assessed.

A major strength is the system's alignment with user needs. The core functionalities—multimodal ingestion, hybrid search, metadata enrichment, fine-grained chunking, and proactive suggestions—directly address pain points such as inefficient retrieval, missed connections, and high cognitive load during synthesis tasks. Because users rely heavily on PDFs, slides, screenshots, and notes across multiple courses and research activities, an application that unifies and semantically indexes this content has clear potential to significantly improve productivity. The solution's emphasis on local processing and offline accessibility further enhances its practicality for TU Wien students, who frequently work on personal devices and may not always have stable connectivity.

From a technical standpoint, the system is challenging but achievable. Many components—OCR, ASR, PDF parsing, embedding-based retrieval, and metadata extraction—are well-supported by existing open-source tools and models. The key innovation lies not in building these components from scratch but in integrating them into a coherent pipeline capable of handling heterogeneous, multi-granular input. Achieving high-quality retrieval will require careful experimentation with chunking strategies, hybrid search architectures, and model selection (especially for bilingual German/English and technical content with formulas). The proactive surfacing of context-sensitive recommendations introduces additional complexity, requiring real-time analysis of user activity without becoming intrusive.

However, there are risks. First, performance may degrade if embeddings or chunking are poorly tuned, leading to irrelevant or noisy search results. Second, multimodal extraction (handwritten text, diagrams, low-quality audio) may produce inconsistent outputs that could harm user trust. Third, the desktop-app architecture, while beneficial for privacy, limits compute resources compared to cloud-based systems. Ensuring efficient local inference, indexing, and storage will require careful optimization.

Evaluation should therefore focus on three dimensions: retrieval quality (precision/recall, relevance), usefulness of generated outputs (summaries, citations, outlines), and user experience (speed, stability, and perceived helpfulness of suggestions). Conducting iterative user testing with students and research teams at TU Wien will be essential to validate whether the system actually reduces search time, increases cross-document connection discovery, and lowers cognitive load during studying and writing.

Overall, the project is ambitious but well-targeted, with strong potential to deliver meaningful improvements in academic knowledge management.

6. Appendix

This appendix outlines several technical directions that can guide the development of our knowledge system. First, the area of retrieval and search focuses on making information access both fast and intelligent. Hybrid search methods that combine keyword matching with semantic understanding can help the system interpret meaning-related terms, while query expansion, embedding models, chunking strategies, and careful preprocessing all contribute to more accurate and relevant results. A second area, proactive surfacing and context-awareness, aims to make the system helpful even before a user asks. By analyzing current activity, it can suggest related materials, cluster topics, identify knowledge gaps, and highlight unexpected connections. The third area, collaborative knowledge management, becomes important when multiple contributors are involved. The system should support merging notes, detecting contradictions, clarifying consensus, tracking authorship, and revealing opportunities for collaboration. A fourth area, understanding and structuring, focuses on transforming unstructured materials into clean, organized knowledge by extracting key concepts, generating metadata, and producing summaries, outlines, or structured study aids. Finally, connection discovery helps uncover meaningful links within the user's materials through semantic similarity, shared entities, temporal patterns, knowledge graphs, and idea-level relationships. Together, these areas provide valuable inspiration for creating a smart, supportive, and connected knowledge platform.